

Denote by $\mathcal{F}_h^{(k)}$ the flow map of the truncated Hamiltonian expansion $\bar{H}_k$,

$$\mathcal{F}_h^{(k)} \stackrel{\text{def}}{=} \exp(h\mathcal{L}_{\bar{H}_k}).$$

By construction, we have

$$\mathbf{z}_{n+1} = \mathcal{F}_h^{(k)}(\mathbf{z}_n) + \boldsymbol{\eta}_n$$

where $\|\boldsymbol{\eta}_n\| \leq Ch^{k+1}$.

Since $\mathcal{F}_h^{(k)}$ preserves its Hamiltonian (3.11), we have

$$\bar{H}_k(\mathcal{F}_h^{(k)}(\mathbf{z}_n)) = \tilde{H}_k(\mathbf{z}_n).$$

Now

$$\begin{aligned} \bar{H}_k(\mathbf{z}_\nu) - \bar{H}_k(\mathbf{z}_0) &= \sum_{n=0}^{\nu-1} \bar{H}_k(\mathbf{z}_{n+1}) - \bar{H}_k(\mathbf{z}_n) \\ &= \sum_{n=0}^{\nu-1} \bar{H}_k(\mathbf{z}_{n+1}) - \tilde{H}_k(\mathcal{F}_h^{(k)}(\mathbf{z}_n)) \\ &= \sum_{n=0}^{\nu-1} \bar{H}_k(\mathcal{F}_h^{(k)}(\mathbf{z}_n) + \boldsymbol{\eta}_n) - \tilde{H}_k(\mathcal{F}_h^{(k)}(\mathbf{z}_n)). \end{aligned}$$

Hence

$$|\bar{H}_k(\mathbf{z}_\nu) - \bar{H}_k(\mathbf{z}_0)| \leq L \sum_{n=0}^{\nu-1} \|\boldsymbol{\eta}_n\| \leq L\nu h^{k+1}.$$

We have $\nu = \tau/h$, thus

$$|\bar{H}_k(\mathbf{z}_\nu) - \bar{H}_k(\mathbf{z}_0)| \leq L\tau h^k.$$

Next observe that

$$H = \bar{H}_k - h^r H_{(r)} - h^{r+1} H_{(r+1)} - \dots - h^k H_{(k)} = \bar{H}_k + \mathcal{O}(h^r).$$

Therefore

$$|H(\mathbf{z}_\nu) - H(\mathbf{z}_0)| \leq L\nu h^{k+1} + \mathcal{O}(h^r),$$